

IBM BOB AI INNOVATION HACKATHON 2026

SupplyGuard AI

AI-Powered Supply Chain Disruption Assistant & Fleet Optimizer

From disruption signals to actionable recovery decisions.

COSMO CODERS

THE PROBLEM

Supply-chain disruptions rarely stay isolated

A disruption can quickly create multiple operational problems:



Shipments become delayed or high-risk



Fleet assets remain idle while capacity is needed elsewhere



Temperature-sensitive shipments can experience unnoticed excursions



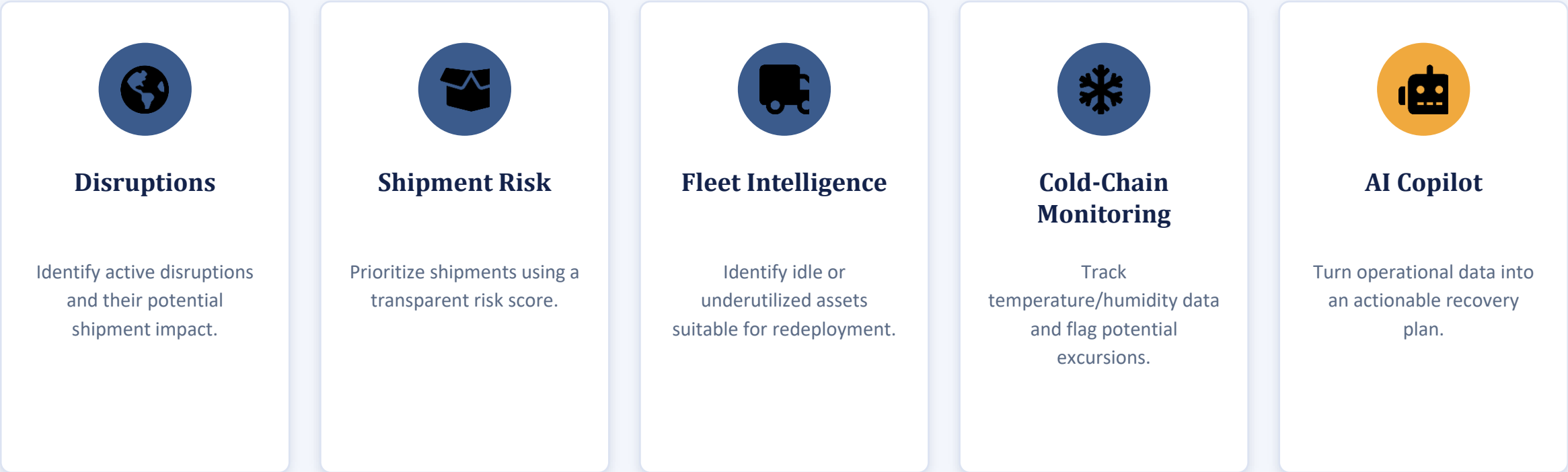
Operations teams must manually connect disruption, shipment and fleet information

THE CHALLENGE

Operators have the data — but need faster visibility into impact, priority and recovery options.

Aligned with the official L2 challenge

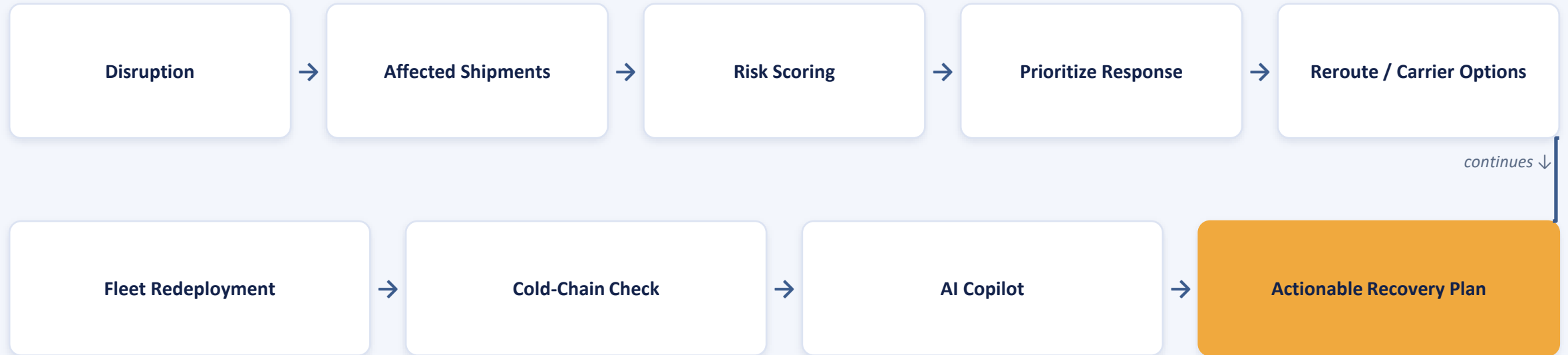
One Control Tower for Disruption Response



BOTTOM LINE



From disruption signal to recovery decision



Signals become a decision

Every stage narrows a raw disruption signal down to one clear, actionable recovery plan — with the operator in control at each step.

Recovery is not just about finding another route

SupplyGuard helps compare recovery alternatives across:

-  **Cost**
What is the operational cost?
-  **Delay**
How much additional time could be introduced?
-  **Risk**
How does the option affect shipment risk?
-  **Emissions**
What is the relative environmental trade-off?

RECOVERY OPTIONS

Option A	<i>Lower cost</i>	Option B	<i>Higher cost</i>	Option C	<i>Balanced trade-off</i>
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Human-in-the-loop decision making

The system provides options and trade-offs — the operator remains in control.

Two operational blind spots. One view.



Fleet Utilisation

- Idle assets
- Available capacity
- Underutilized vehicles
- Refrigerated vehicles
- Location and capacity information

Goal: identify assets that could support recovery.



Cold Chain

- Temperature
- Humidity
- Shipment sensor timeline
- Warning / critical excursions

Early visibility helps operators investigate temperature-sensitive shipments before delivery.

Note: monitoring, not regulatory certification.

Ask the system what to do next

"What should I do about SHP-104?"

SUPPLYGUARD AI RESPONDS WITH

- ✓ Current shipment risk
- ✓ Disruption impact
- ✓ Relevant fleet options
- ✓ Key risk factors
- ✓ Recommended recovery actions
- ✓ Cold-chain considerations

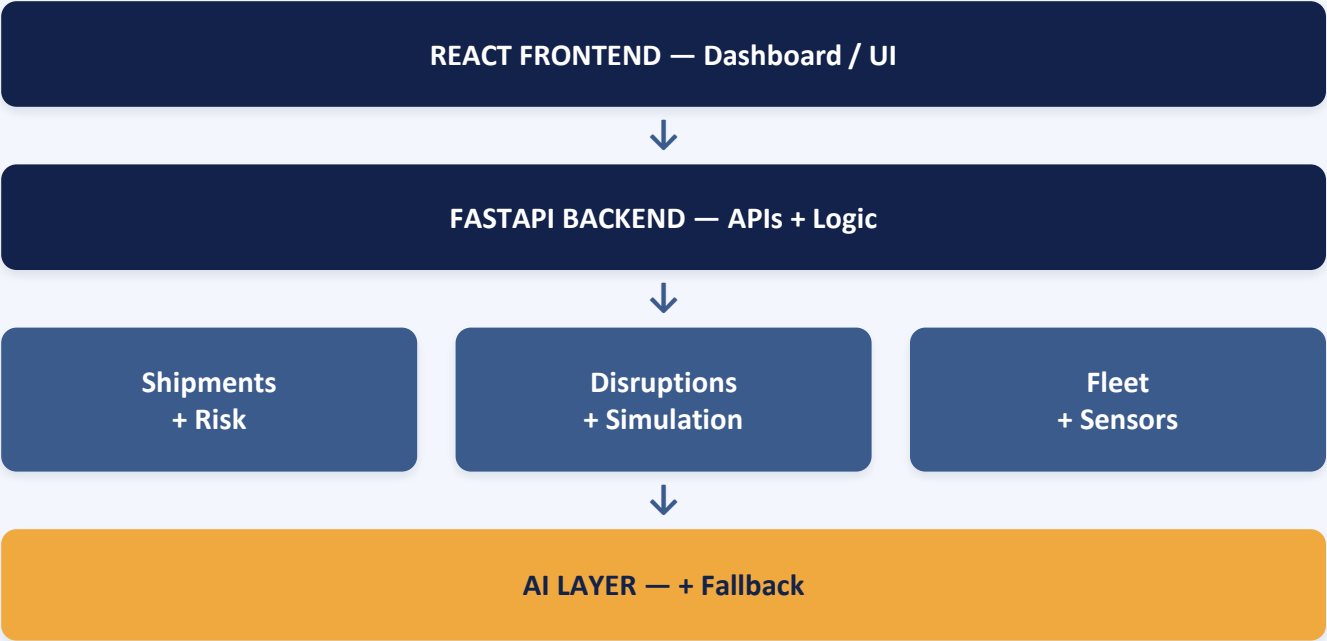


Natural language, not multiple screens

Instead of searching across multiple screens, operators can ask questions in natural language.

A very good demo moment

Built as a practical, local-first prototype





Frontend

React • Vite • Tailwind CSS • Recharts



Backend

Python • FastAPI • Pydantic



Data

SQLite • Structured prototype datasets

Only technologies actually present in the final repository are listed.

How IBM Bob supported development



Repository Analysis

Understanding the existing repository



Architecture & Planning

Architecture and implementation planning



Development Assistance

Hands-on development assistance



Debugging & Iteration

Debugging and iteration



Code Review

Code review / validation



Documentation

Supporting documentation and project structure

KEY MESSAGE

Bob was integrated into the development workflow — not simply mentioned in the presentation.

From disruption to recovery

01

Dashboard

See the operational situation.

02

Simulate disruption

Create a disruption scenario.

03

Identify affected shipments

See which shipments are impacted.

04

Investigate critical shipment

Understand the risk and contributing factors.

05

Evaluate recovery

Compare rerouting / fleet options.

06

Check cold chain

Inspect temperature-related alerts.

07

Ask AI Copilot

"What should I do about SHP-104?"

08

Get recovery guidance

Visibility → Prioritisation → Recovery.

What SupplyGuard enables

TODAY — PROTOTYPE

- ✓ Unified operational visibility
- ✓ Disruption simulation
- ✓ Shipment risk prioritisation
- ✓ Fleet utilisation analysis
- ✓ Cold-chain monitoring
- ✓ AI-assisted operational recommendations

FUTURE SCOPE

- Live disruption feeds
- GPS / IoT integrations
- Real-time route optimisation
- ERP / TMS integration
- Predictive disruption detection
- Enterprise-scale deployment

Future capabilities are intentionally separated from what is implemented today.

SupplyGuard AI

Turning supply-chain uncertainty into actionable decisions.

Disruption Intelligence × Shipment Risk × Fleet Optimisation × Cold-Chain Monitoring × AI Decision Support

COSMO CODERS

IBM Bob AI Innovation Hackathon 2026

Thank You